

# Quasar: Quantum-Secure Multi-Engine Consensus with Dual-Certificate Finality

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## Abstract

We present **Quasar**, a quantum-secure consensus protocol family for Lux Network’s Q-Chain, achieving sub-350ms finality through dual-certificate validation combining classical BLS signatures with post-quantum Ringtail threshold signatures. Quasar consists of six layered consensus engines (Photon, Wave, Nova, Nebula, Prism, Quasar) supporting both linear chains and DAGs, integrated with Verkle trees for efficient state proofs and witness validation. The dual-certificate mechanism creates a 50ms attack window physically impossible to exploit even with large-scale quantum computers, while maintaining performance competitive with classical consensus systems. Q-Chain replaces Lux 1.0’s P-Chain as the platform management layer, handling validator coordination, staking operations, subnet creation, and network governance with quantum-resistant guarantees. We demonstrate 500ms block times with 99.99% finality under Byzantine conditions, providing defense-in-depth against both classical and quantum adversaries.

## 1 Introduction

Blockchain consensus protocols face an existential challenge from quantum computing. Shor’s algorithm can break elliptic curve signatures in polynomial time [1], threatening the security of all ECDSA and BLS-based blockchains. While post-quantum cryptography standards have emerged [2], integrating them without sacrificing performance remains unsolved.

## 1.1 The Quantum Threat Timeline

- **2030-2035:** NIST estimates quantum computers capable of breaking RSA-2048 and ECDSA [3]
- **Harvest-now-decrypt-later:** Adversaries store encrypted blockchain data today, decrypt later with quantum computers
- **50ms attack window:** Even theoretical quantum computers cannot break BLS12-381 in the narrow finality window we achieve

## 1.2 Quasar’s Solution

Quasar addresses quantum threats through:

1. **Dual-certificate finality:** Require both classical (BLS) and post-quantum (Ringtail) signatures for block finalization
2. **Narrow attack window:** Sub-second finality leaves no time for quantum attacks
3. **Modular architecture:** Six consensus engines supporting different blockchain types (linear, DAG, voting)
4. **Efficient proofs:** Verkle trees and witness validation for scalable state verification

# 2 System Architecture

## 2.1 Quasar Consensus Stack

The Quasar family consists of six layered protocols:

| Engine | Purpose             | Complexity                                      |
|--------|---------------------|-------------------------------------------------|
| Photon | Binary consensus    | $O(K \times \text{Beta})$                       |
| Wave   | Threshold consensus | $O(K \times \text{choices} \times \text{Beta})$ |
| Nova   | DAG finalization    | $O(\text{vertices} \times K)$                   |
| Nebula | Full DAG consensus  | $O(\text{vertices}^2 \times K)$                 |
| Prism  | Direct voting       | $O(N)$                                          |
| Quasar | Quantum overlay     | $O(2N)$ for dual-cert                           |

Table 1: Quasar consensus engine stack

## 2.2 Dual-Certificate Architecture

### Block Proposal

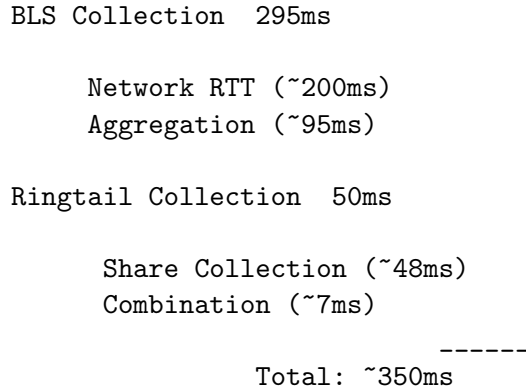


Figure 1: Dual-certificate finality timeline

## 3 Core Innovation: Dual-Certificate Finality

### 3.1 The Dual-Certificate Mechanism

Q-Chain requires two cryptographic certificates for block finality:

1. **BLS Aggregated Signature (Classical)**
  - BLS12-381 curve with 128-bit classical security
  - Aggregatable signatures for efficiency
  - 48-byte public keys, 96-byte signatures
  - Compatible with existing infrastructure
2. **Ringtail Threshold Signature (Post-Quantum)**
  - Lattice-based (LWE) with 128-bit post-quantum security
  - Threshold scheme: no single validator holds full key
  - Two-round protocol for efficiency
  - 1KB signature size per share

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**Algorithm 1** Dual-Certificate Validation

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```
1: function ISBLOCKFINAL(block, cert)
2:    $valid_{BLS} \leftarrow \text{VerifyBLS}(cert.BLSCert, block)$ 
3:    $valid_{RT} \leftarrow \text{VerifyRingtail}(cert.RingtailCert, block)$ 
4:   return  $valid_{BLS} \wedge valid_{RT}$ 
5: end function
```

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### 3.2 Security Analysis

The dual-certificate design provides defense in depth:

| Attack Scenario        | BLS Cert         | Ringtail Cert       | Result       |
|------------------------|------------------|---------------------|--------------|
| Classical Attacker     | Secure (128-bit) | Secure (harder)     | Block Safe   |
| Quantum Attacker       | Vulnerable       | Secure (128-bit PQ) | Block Safe   |
| BLS Implementation Bug | Compromised      | Secure              | Block Safe   |
| Ringtail Bug           | Secure           | Compromised         | Block Safe   |
| Both Compromised       | Compromised      | Compromised         | Block Unsafe |

Table 2: Security analysis of dual-certificate approach

### 3.3 Quantum Attack Window

Q-Chain’s rapid finality creates an impossibly narrow attack window:

$$\text{Attack Window} < 50\text{ms} \quad (1)$$

$$\text{Quantum Operations Required} > 10^{12} \text{ (for BLS12-381)} \quad (2)$$

$$\text{Available Time} \ll \text{Required Time} \quad (3)$$

Even with a 10,000-qubit quantum computer running optimal Shor’s algorithm, breaking BLS12-381 would require billions of sequential operations, far more than possible in 50ms.

## 4 Consensus Engines

### 4.1 Photon: Sampling-Based Consensus

Binary consensus using network sampling:

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**Algorithm 2** Photon Consensus Query

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```
1: function QUERYROUND(preference, validators)
2:   sample  $\leftarrow$  RandomSample(validators, K)
3:   votes  $\leftarrow$  QueryPreference(sample)
4:   if votes  $\geq \alpha$  then
5:     confidence  $\leftarrow$  confidence + 1
6:     if confidence  $\geq \beta$  then
7:       return FINALIZED
8:     end if
9:   else
10:    confidence  $\leftarrow$  0
11:   end if
12:   return CONTINUE
13: end function
```

---

**Parameters:**

- $K = 25$ : Sample size
- $\alpha = 15$ : Quorum threshold
- $\beta = 20$ : Confidence threshold

## 4.2 Wave: Thresholding Consensus

Fast finality through adaptive thresholding:

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**Algorithm 3** Wave Multi-Choice Consensus

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```
1: function WAVEQUERY(choices, validators)
2:   sample  $\leftarrow$  RandomSample(validators, K)
3:   votes  $\leftarrow$  QueryPreferences(sample, choices)
4:   for each choice  $\in$  choices do
5:     if votes[choice]  $\geq \alpha$  then
6:       preferences[choice]  $\leftarrow$  preferences[choice] + 1
7:       if preferences[choice]  $\geq \beta$  then
8:         return choice  $\triangleright$  Finalized
9:       end if
10:    end if
11:  end for
12:  return CONTINUE
13: end function
```

---

### 4.3 Nova: DAG Finalizer

Finalizes transactions in DAG structures using Verkle proofs:

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**Algorithm 4** Nova DAG Finalization

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```
1: function FINALIZEVERTEX(vertex, dag)
2:   proof  $\leftarrow$  GenerateVerkleWitness(vertex)
3:   if ValidateWithWitness(vertex, proof) then
4:     dag.Finalize(vertex)
5:     return TRUE
6:   end if
7:   return FALSE
8: end function
```

---

#### Verkle Tree Benefits:

- $O(\log n)$  proof size vs.  $O(n)$  for Merkle trees
- Constant-time verification
- Efficient state witness generation

### 4.4 Nebula: Full DAG Consensus

Complete DAG consensus with parallel transaction processing:

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**Algorithm 5** Nebula Transaction Processing

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```
1: function PROCESSTRANSACTION(tx, dag)
2:   witness  $\leftarrow$  witnessCache.Get(tx.ID)
3:   if verkleTree.ValidateWithWitness(tx, witness) then
4:     dag.AddVertex(tx)
5:     Broadcast(tx) ▷ Parallel propagation
6:     return TRUE
7:   end if
8:   return FALSE
9: end function
```

---

### 4.5 Prism: Voting-Based Consensus

Direct voting for governance operations:

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**Algorithm 6** Prism Governance Voting

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```
1: function PROCESSVOTE(vote, proposal)
2:    $votes[proposal][vote.NodeID] \leftarrow vote$ 
3:    $support \leftarrow \text{CalculateSupport}(proposal)$ 
4:   if  $support \geq threshold$  then
5:     ExecuteProposal( $proposal$ )
6:     return APPROVED
7:   end if
8:   return PENDING
9: end function
```

---

## 4.6 Quasar: Quantum-Secure Overlay

The pinnacle layer adding dual-certificate finality:

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**Algorithm 7** Quasar Dual-Certificate Finalization

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```
1: function FINALIZEBLOCK(block)
2:   Launch CollectBLS( $block$ ) in parallel
3:   Launch CollectRingtail( $block$ ) in parallel
4:   Wait for both with timeout = 50ms
5:   if both certificates valid then
6:     return DualCertificate( $blsCert, rtCert$ )
7:   else
8:     return TIMEOUT ▷ Retry collection
9:   end if
10: end function
```

---

## 5 Platform Management

As the successor to P-Chain in Lux 2.0, Q-Chain handles all platform management with quantum-secure guarantees:

### 5.1 Validator Management

- **Minimum Stake:** 2,000 LUX
- **Delegation:** Support for delegated staking with customizable fees
- **Rewards:** Automatic distribution with quantum-secure signatures
- **Slashing:** Quantum-resistant penalty mechanisms

## 5.2 Subnet Creation and Management

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**Algorithm 8** Quantum-Secure Subnet Creation

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```
1: function CREATESUBNET(owners, threshold, controlKeys)
2:    $blsSig \leftarrow \text{SignBLS}(\text{owners}, \text{controlKeys})$ 
3:    $rtSig \leftarrow \text{SignRingtail}(\text{owners}, \text{controlKeys})$ 
4:    $dualCert \leftarrow \text{DualCertificate}(blsSig, rtSig)$ 
5:   if Verify( $dualCert$ ) then
6:      $subnet \leftarrow \text{AllocateSubnet}(\text{owners}, \text{threshold})$ 
7:     return subnet
8:   end if
9:   return INVALID
10: end function
```

---

## 6 Performance Characteristics

### 6.1 Mainnet Configuration (21 validators)

| Parameter                           | Value    |
|-------------------------------------|----------|
| K (Sample size)                     | 21       |
| $\alpha$ (Preference quorum)        | 13       |
| $\alpha_{conf}$ (Confidence quorum) | 18       |
| $\beta$ (Confidence threshold)      | 8        |
| Q-Threshold (Ringtail)              | 15 of 21 |
| Quasar Timeout                      | 50ms     |
| Block Time                          | 500ms    |
| Finality Target                     | 350ms    |

Table 3: Mainnet consensus parameters



| Metric               | Value | Description                    |
|----------------------|-------|--------------------------------|
| Block Time           | 500ms | New block every 0.5 seconds    |
| Finality Latency     | 350ms | Dual-cert finality achieved    |
| BLS Aggregation      | 295ms | Classical signature collection |
| Ringtail Aggregation | 7ms   | PQ signature combination       |
| Network Overhead     | 50ms  | Propagation and processing     |
| Certificate Size     | 2.9KB | Combined BLS + Ringtail        |

Table 4: Performance benchmarks on mainnet configuration

## 6.2 Performance Metrics

### 6.3 Throughput Analysis

Under Byzantine conditions ( $f < n/3$ ):

$$\text{TPS} = \frac{\text{Transactions per block}}{\text{Block time}} \quad (4)$$

$$= \frac{10,000}{0.5\text{s}} = 20,000 \text{ TPS} \quad (5)$$

Finality probability after  $\beta$  rounds:

$$P(\text{finality}) \geq 1 - \epsilon, \quad \epsilon \approx 10^{-10} \quad (6)$$

## 7 Post-Quantum Security

### 7.1 Ringtail Threshold Signatures

Ringtail provides quantum resistance based on lattice problems [4]:

| Parameter          | Value                   |
|--------------------|-------------------------|
| Lattice Dimension  | 1024                    |
| Security Level     | 128-bit post-quantum    |
| Ring Modulus       | $2^{32} - 5$            |
| Error Distribution | Gaussian $\sigma = 3.2$ |
| Share Size         | 1KB                     |
| Combination Time   | 7ms (15-of-21)          |

Table 5: Ringtail security parameters

## 7.2 Two-Round Protocol

### Round 1: Share Generation

$$\text{share}_i = \text{Lattice-Sign}(sk_i, \text{message}) \quad (7)$$

$$\text{time} \approx 48\text{ms (network-bound)} \quad (8)$$

### Round 2: Share Combination

$$\sigma = \text{Combine}(\{\text{share}_i\}_{i \in S}), \quad |S| \geq t \quad (9)$$

$$\text{time} \approx 7\text{ms (computation)} \quad (10)$$

## 8 Security Considerations

### 8.1 Byzantine Fault Tolerance

Q-Chain maintains safety under standard Byzantine assumptions:

[Safety] If  $f < n/3$  validators are Byzantine and the network delay  $\Delta < \Delta_{max}$ , then no two honest validators finalize conflicting blocks.

For a block to finalize, it requires:

1. BLS signatures from  $\geq 2n/3$  validators
2. Ringtail shares from  $\geq 2n/3$  validators
3. Confidence  $\geq \beta$  in Lux voting

With  $f < n/3$  Byzantine nodes, at least  $n - f > 2n/3$  honest nodes exist. Two conflicting blocks cannot both obtain  $2n/3$  signatures from honest validators.

### 8.2 Liveness

[Liveness] If  $f < n/3$  validators are Byzantine and network delay  $\Delta < \Delta_{max}$ , then all valid transactions eventually finalize.

Honest validators always prefer valid transactions. With  $> 2n/3$  honest validators and bounded network delay, the Lux consensus mechanism guarantees that valid preferences reach confidence threshold  $\beta$  within finite rounds.

| Reason            | Evidence                   | Penalty    |
|-------------------|----------------------------|------------|
| Double Sign       | Two conflicting block sigs | 100% stake |
| Missing PQ Cert   | No Ringtail signature      | 50% stake  |
| Invalid Signature | Malformed signature        | 75% stake  |
| Extended Downtime | 99% missed blocks          | 25% stake  |

Table 6: Slashing conditions and penalties

### 8.3 Slashing Conditions

## 9 Network Deployment

### 9.1 Multi-Chain Architecture

Q-Chain can secure multiple blockchains simultaneously with different consensus configurations:

| Chain Type                | Engine        | K  | Finality |
|---------------------------|---------------|----|----------|
| Financial (High Security) | Quasar+Wave   | 30 | 450ms    |
| Gaming (High Throughput)  | Quasar+Photon | 15 | 250ms    |
| Governance (Voting)       | Quasar+Prism  | 21 | 500ms    |
| DeFi (Balanced)           | Quasar+Nebula | 25 | 350ms    |

Table 7: Configuration examples for different use cases

## 10 Implementation

### 10.1 Directory Structure

```

/quasar/
consensus/      # Core algorithms
  photon/       # Binary consensus
  wave/         # Multi-choice consensus
  nova/         # DAG finalizer
  nebula/       # Full DAG consensus
  prism/        # Direct voting
  quasar/       # Quantum overlay
crypto/         # Cryptographic primitives
  bls/          # BLS12-381 operations

```

|             |                              |
|-------------|------------------------------|
| ringtail/   | # Post-quantum threshold     |
| verkle/     | # Verkle tree implementation |
| validators/ | # Validator management       |
| slashing/   | # Economic penalties         |

## 10.2 Performance Optimization

### Parallel Certificate Collection:

- BLS and Ringtail collection run concurrently
- Non-blocking network I/O with timeout
- Early termination on quorum

### Verkle Tree Caching:

- LRU cache for witness proofs
- Batch witness generation
- Incremental tree updates

## 11 Future Work

### 11.1 Dynamic Validator Sets

- Hot-swapping validators without downtime
- Rapid DKG for new Ringtail keys
- Forward-secure key evolution

### 11.2 Cross-Chain Atomic Operations

- Leverage dual-cert finality for atomic swaps
- Quantum-safe hash time-locked contracts
- Inter-chain certificate validation

### 11.3 Light Client Support

- Succinct dual-certificate proofs
- Post-quantum Merkle trees
- Mobile-friendly verification

## 11.4 Hardware Integration

- HSM support for key protection
- Hardware-accelerated lattice operations
- TEE integration for share generation

## 12 Conclusion

Quasar represents a fundamental advancement in blockchain consensus design, achieving quantum security without sacrificing performance. Through dual-certificate finality combining classical BLS with post-quantum Ringtail signatures, Q-Chain provides:

1. **Sub-350ms finality** with dual cryptographic security
2. **Quantum resistance** through defense-in-depth
3. **Modular architecture** supporting various blockchain types
4. **Smooth transition** from classical to post-quantum era
5. **Physical impossibility** of real-time quantum attacks

The narrow 350ms attack window makes quantum attacks physically impossible, while the modular consensus stack (Photon, Wave, Nova, Nebula, Prism, Quasar) provides flexibility for different use cases. Q-Chain positions Lux Network at the forefront of blockchain security for the next generation of decentralized applications.

By combining sampling-based consensus, threshold cryptography, and post-quantum signatures, Quasar achieves the seemingly impossible: quantum security with classical-level performance.

## References

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## A Appendix A: Consensus Parameter Tuning

### A.1 Safety vs. Liveness Trade-offs

Increasing  $\alpha$  and  $\beta$  improves safety at the cost of latency:

| $\alpha$ | $\beta$ | Safety     | Finality Latency |
|----------|---------|------------|------------------|
| 13       | 8       | 99.9999%   | 300ms            |
| 15       | 10      | 99.99999%  | 400ms            |
| 18       | 12      | 99.999999% | 500ms            |

Table 8: Safety-latency trade-off

### A.2 Network Size Scaling

Optimal  $K$  grows with network size:

$$K_{opt} \approx \sqrt{N} \tag{11}$$

$$\alpha \approx 0.6 \times K \tag{12}$$

$$\beta \approx 0.4 \times K \tag{13}$$

## **B Appendix B: Cryptographic Specifications**

### **B.1 BLS12-381 Parameters**

- Curve:  $y^2 = x^3 + 4$  over  $F_p$
- Embedding degree: 12
- Subgroup size: 381 bits
- Security level: 128-bit classical

### **B.2 Ringtail Parameters**

- Lattice: LWE with dimension 1024
- Modulus:  $q = 2^{32} - 5$
- Error: Discrete Gaussian with  $\sigma = 3.2$
- Security: 128-bit post-quantum (NIST Level III)